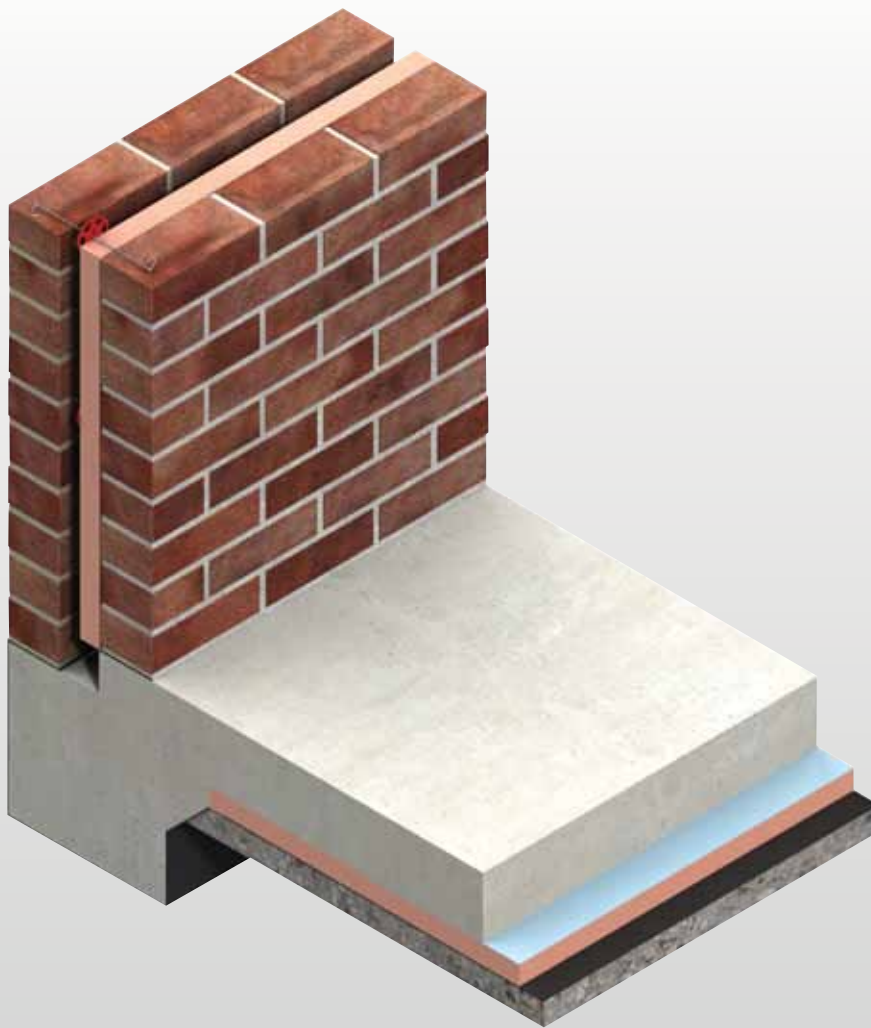




Kooltherm® K3 Floorboard

INSULATION FOR FLOORS



- Super high performance rigid thermoset phenolic insulation
- Fibre-free, closed cell insulation core
- Resistant to the passage of water vapour
- Easy to handle and install
- Ideal for new build and refurbishment
- Manufactured with a CFC/HCFC-free blowing agent that has zero ODP and low GWP
- Compliant with AS/NZS 4859.1



*Low Energy –
Low Carbon Buildings*

Typical Construction and Total Construction R-values

Solid Concrete Ground Based Floors - Insulation below the Floor Slab

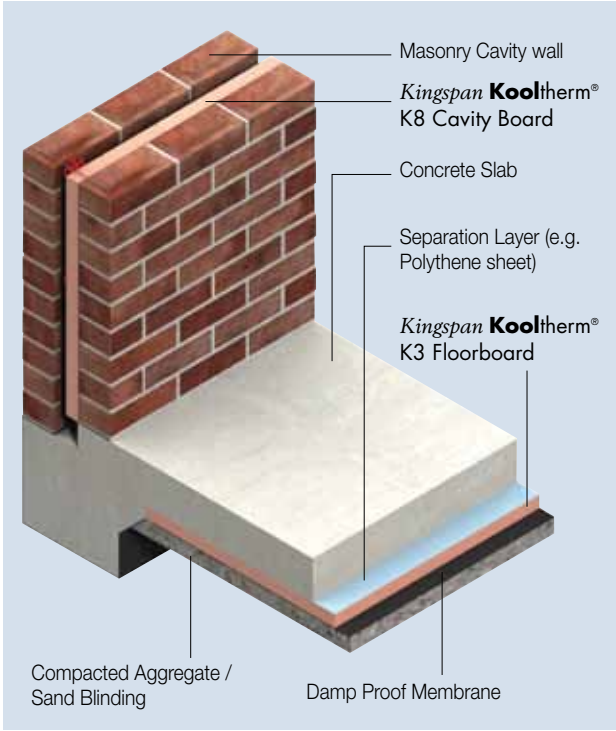


Figure 1

Total R-values for various thicknesses of <i>Kingspan Kooltherm</i> ® K3 Floorboard	
Product Thickness	Heat flow out
25 mm	R _T 1.5
30 mm	R _T 1.7
40 mm	R _T 2.2
50 mm	R _T 2.8

Assumptions

The above thermal calculations are taken through the construction profile with 150 mm concrete slab and *Kingspan Kooltherm*® K3 Floorboard.

Kingspan Kooltherm® products are manufactured, tested and packaged in conformance with AS/NZS 4859.1.

The contribution of the product Total R-value depends on installation and environmental conditions.

Product Details

Product Description

Kingspan Kooltherm® K3 Floorboard is a super high performance, fibre-free rigid thermoset closed cell phenolic insulation core, sandwiched between an upper tissue-based facing and a lower facing of highly reflective aluminium foil autohesively bonded to the insulation core during manufacture.



Kingspan Kooltherm® K3 is manufactured without the use of CFCs/HCFCs and has zero Ozone Depletion Potential (ODP) and low Global Warming Potential (GWP).



Product Data

Thermal Conductivity (λ -value)	0.020 W/m K (Insulant thickness ≥ 45 mm) 0.021 W/m K (Insulant thickness 25 - 44 mm)
Product Dimensions	2270 mm x 1200 mm (2.72 m ²)
Product Thickness	25, 30, 40, 50 mm

Product R-values

Product Thickness	Product R-value
25 mm	R1.2
30 mm	R1.4
40 mm	R1.9
50 mm	R2.5

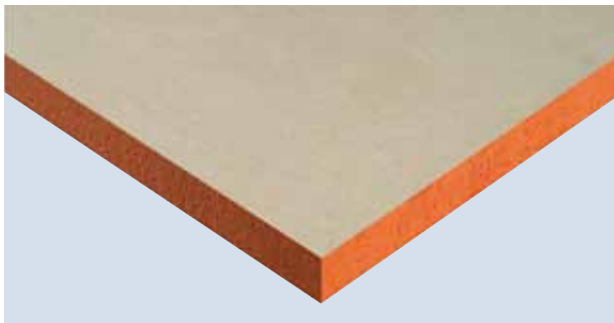


Figure 2 Super high performance *Kingspan Kooltherm*® K3 Floorboard

Specification Guide

Kingspan Kooltherm® K3 Floorboard

The floor insulation shall be *Kingspan Kooltherm*® K3 Floorboard ____ mm thick comprising a CFC/HCFC-free and zero Ozone Depletion Potential (ODP) rigid thermoset phenolic insulation core with a tissue based facing on one side and composite foil on the other side, manufactured under a management system certified to BS / I.S. EN ISO 9001:2008, BS / I.S. EN ISO 14001:2004 and BS / I.S. OHSAS 18001:2007 by Kingspan Insulation Limited and shall be installed in accordance with the instructions issued by them.

Compressive Loads

Un-reinforced floor screeds can be used in conjunction with *Kingspan Kooltherm*® K3 Floorboard in most applications. However, where floor loads are to be excessive, please contact our Technical Services Department on 1300 247 235 or email technical@kingspaninsulation.com.au.

Substrate

Kingspan Kooltherm® K3 Floorboard is not recommended for use in direct contact with subsoil and must be used over a DPM. Where vertical slab edge insulation is required, the DPM should always be placed on the external face of the *Kingspan Kooltherm*® K3 Floorboard.

Wheeled / Foot Traffic

Ensure boards are protected during installation from wheeled / foot traffic by using scaffold planks or other protective measures.

Standards and Approvals

Kingspan Kooltherm® K3 Floorboard is manufactured to the highest standards and certified under the following management systems:

Standard	Management System
BS / I.S. EN ISO 9001:2008	Quality Management
BS / I.S. EN ISO 14001:2004	Environmental Management
BS / I.S. OHSAS 18001:2007	Health and Safety Management

Product Testing

Characteristic	Standard	Result
Compressive Strength	BS EN 826:1996	Typically exceeds 100 kPa at 10% compression
Water Vapour Resistance	BS EN 12086:1997 / I.S. EN 12086:1998	> 100 MN-s/g

Fire Performance

Test	Test Method	Result
Ignitability, Flame spread, Heat release, Smoke release	AS 1530.3	0/0/0/0-1
Fire Propagation	BS 476 Part 6	Class 0* (in accordance with the Building Regulations in England, Wales and Northern Ireland)
Surface Spread of Flame	BS 476 Part 7	

* Applies only to the *Kingspan Kooltherm*® insulation core.

Durability

If correctly applied, *Kingspan **Kooltherm***[®] products can be expected to have a long life of service.

Their durability depends on the supporting structure and the conditions of its use.

*Kingspan **Kooltherm***[®] products are warranted for a period of 10 years for both residential and commercial installations.*

* Subject to the terms of the complete *Kingspan **Kooltherm***[®] warranty document which is available upon request or downloadable from www.kingspaninsulation.com.au.

Environmental Data

Aspect	Characteristic
Recyclability	Non-contaminated insulation site waste is recyclable, but there are currently no facilities in Australia to process returned material
Re-usability	Re-usable if removed with care (long term of service expected)
Water Use	No water used in Kingspan Insulation's manufacturing process
Blowing Agent Global Warming Potential (GWP)	Manufactured with a blowing agent that has low GWP
Blowing Agent Ozone Depletion Potential (ODP)	Manufactured with a CFC/HCFC-free blowing agent that has zero ODP
Packaging	Contains 0% recycled product Polythene wrap and EPS skids 100% recyclable

Installation Instructions

1. The site should be prepared and foundations, where appropriate, built to damp proof course (DPC) level.
2. A thin sand blinding may be used to achieve a continuous level surface free from projections over rolled aggregate.
3. The damp proof membrane (minimum 300 micron / 1200 gauge polythene) should be laid with joints well lapped and folded, to prevent the passage of ground water, over well compacted aggregate, prior to laying **Kingspan Kooltherm® K3 Floorboard**.
4. The membrane should be brought up the surrounding foundation walls/slab until it is sufficiently above the height of the wall DPC so that it will connect with or form the DPC.
5. **Kingspan Kooltherm® K3 Floorboard** should not be placed under any direct point loads or foundations including slab thickening.
6. The insulation boards should always be loose-laid break-bonded, with joints lightly butted.
7. If two layers of **Kingspan Kooltherm® K3 Floorboard** are required, they should be horizontally offset relative to each other so that, as far as possible, the board joints in the two adjacent layers do not coincide with each other. To avoid glare, lay with the foil face facing down.
8. Insulation boards should be overlaid with a polythene sheet (not less than 125 micron / 500 gauge) as best practice, to prevent the wet concrete penetrating the joints between the boards, and to act as a vapour control layer. Ensure the polythene sheet has 150 mm overlaps, taped at the joints.
9. The subsequent installation of the concrete slab and screed or other flooring material is carried out in a manner similar to that for an un-insulated floor. The concrete slab and screed should be allowed to dry out prior to the installation of the floor finish.

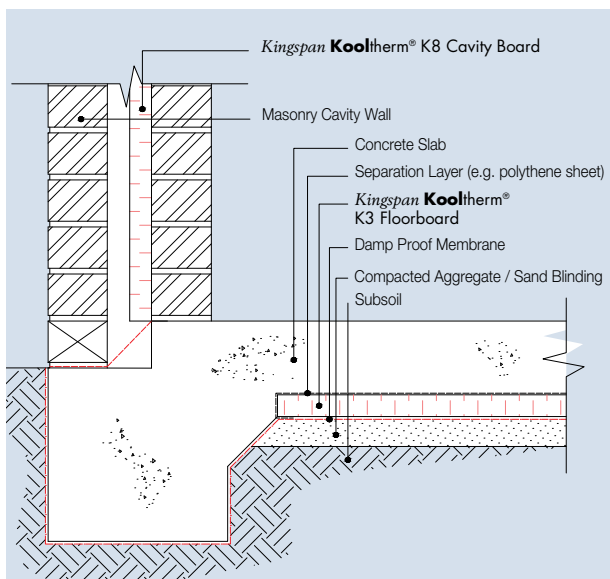


Figure 3 Side elevation - Cavity Wall on Ground Floor Slab with **Kingspan Kooltherm® K3 Floorboard**

General Requirements

Cutting

Cutting should be carried out either by using a fine toothed saw, or by scoring with a sharp knife, snapping the board over a straight edge and then cutting the facing on the other side. Ensure accurate trimming to achieve close-butting joints and continuity of insulation.

Packaging

According to quantity, the boards are supplied in packs, labelled and shrink-wrapped in polythene.

Handling and Storage

The polythene packaging of Kingspan Kooltherm® should not be considered adequate for long term outdoor protection. Ideally, boards should be stored inside a building. If, however, outside storage cannot be avoided, then the boards should be stacked clear of the ground and covered with a polythene sheet or weatherproof tarpaulin. Boards that have been allowed to get wet should not be used.

Resistance to Solvents

The insulation core is resistant to short-term contact with petrol and with most dilute acids, alkalis and mineral oils. However, it is recommended that any spills be cleaned off fully before the boards are installed. Ensure that safe methods of cleaning are used, as recommended by suppliers of the spilt liquid. The insulation core is not resistant to some solvent-based adhesive systems, particularly those containing methyl ethyl ketone. Adhesives containing such solvents should not be used in association with this product. Damaged boards or boards that have been in contact with harsh solvents or acids should not be used.

OH & S

Kingspan Insulation products are chemically inert and safe to use. A Product Safety Information sheet is available from Kingspan Insulation Pty Ltd.

Please note that the reflective surfaces on this product are designed to enhance their thermal performance. As such, they will reflect light as well as heat, including ultraviolet light. Therefore, if these boards are being installed during bright or sunny weather, it is advisable to wear UV protective sunglasses or goggles and if the skin is exposed for a significant period of time, to protect bare skin with a UV block sun cream.

Foil facings are conductive to electricity - avoid contact with un-insulated electrical cables and fittings.

Contact Details

General Enquiries

Tel: 1300 247 235

Email: info@kingspaninsulation.com.au

Kingspan Insulation Pty. Ltd. reserves the right to amend product specifications without prior notice. The information, technical details and fixing instructions etc. included in this literature are given in good faith and apply to uses described. Recommendations for use should be verified as to the suitability and compliance with actual requirements, specifications and any applicable laws and regulations. For other applications or conditions of use, Kingspan Insulation offers a Technical Advisory Service the advice of which should be sought for uses of Kingspan Insulation products that are not specifically described herein. Please check that your copy of the literature is current by contacting us or visiting www.kingspaninsulation.com.au



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